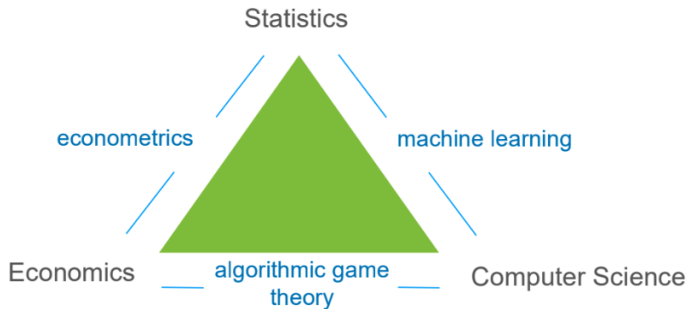


Методы автоматического выставления ставок в рекламных аукционах

Andrey Pudovikov

Some Academic Foundations



Motivation: Online Advertising

«Купить самсунг»: объявления для Москвы 12 027

Все категории
Электроника
Телефоны
Мобильные телефоны

Производитель

Samsung

Модель

Модель

Память

- До 4 ГБ
- 4 ГБ
- 8 ГБ
- 16 ГБ
- 32 ГБ

Показать ещё

Оперативная память

- До 0.5 ГБ
- 0.5 ГБ
- 1 ГБ
- 1.5 ГБ
- 2 ГБ

Показать ещё

SIM-карты

- SIM + eSIM
- 2 SIM
- 1 SIM
- eSIM



11 Сортировка

Сначала из Москвы

Уведомлять о новых



Samsung Galaxy S10e, 8/128 ГБ
13 000 Р

Скидка на доставку 15 дней на возврат

Новый

с/д Доставка от 1 дня

2sim. Запечатанный. Гарантия. Комплект: Переходник Usb type C — Usb. Наушники Akg. Кабель Usb. Сетевое зарядное устройство. Скрепка для извлечения Sim-карты. Документация. Доставка курьером по Москве и МО. Отправляем Любыми ТК п...

Продвинуто



SVOIOPT

★★★★★ 4,4 · 450 отзывов

Документы проверены



Samsung Galaxy A56, 12/256 ГБ
27 390 Р

Рыночная цена

Новый

Курская, 6–10 мин.

Нашли дешевле? Пишите или звоните Сейчас и мы сделаем лучшую цену. При покупке сегодня: Подарок или Скидка! Гарантия: до 24 месяцев. Акция Trade-IN: обменный смартфон вышл...

Продвинуто



BAZA

★★★★★ 5,0 · 13 577 отзывов

Реквизиты проверены

Показать телефон

Написать



Samsung Galaxy S25, 12/256 ГБ
51 490 Р

Цена ниже рыночной

Новый

Курская, 6–10 мин.

Нашли дешевле? Напишите нам, постараемся сделать лучшее предложение! При покупке сегодня: Подарок или Скидка! Гарантия: до 24 месяцев. Акция Trade-IN: обменный смартфон вышл...

Продвинуто



BAZA

★★★★★ 5,0 · 13 577 отзывов

Реквизиты проверены

Показать телефон

Написать

The recommendation process

Многостадийные рексистемы

Рекомендации – многостадийный процесс:

- На ранних стадиях – более быстрые и эффективные алгоритмы
- Дальше – более тяжелые модели



RTB vs Autobidding

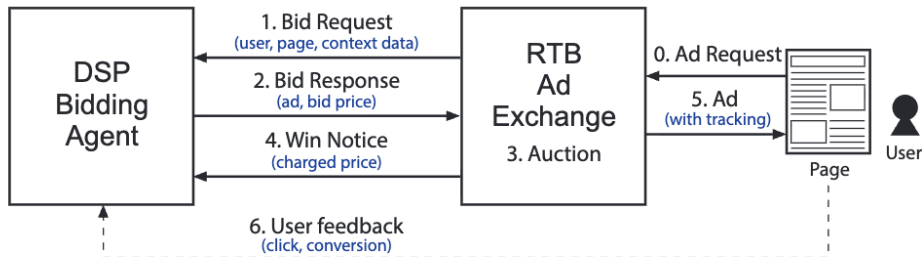


Figure 1: A brief illustration of the interactions between user, ad exchange and DSP.

RTB computes a bid for each impression in real time. [8]

RTB vs Autobidding

Real-Time Bidding	Autobidding
Time scale ~100 ms	Time scale Slightly more
Responsibility One bid per impression	Responsibility May design the whole system

Autobidding problem

Optimisation, dual variables, and interpretation

Auction format

1. Single-slot First Price (Strategy - bid shading)
2. **Single-slot Second Price**
3. Generalized Second Price (GSP)
4. Vickrey–Clarke–Groves auction (VCG)

Autobidding Problem Formulation (SPA)

$$\begin{aligned} \max_{x_t} \quad & \sum_{t=1}^T x_t \cdot CTR_t \cdot CVR_t \\ \text{s.t.} \quad & \sum_{t=1}^T x_t \cdot wp_t \leq B \\ & \frac{\sum_{t=1}^T x_t \cdot wp_t}{\sum_{t=1}^T x_t \cdot CTR_t} \leq C \end{aligned}$$

Where x_t is a binary variable, equals to 1 iff bidder wins impression t . wp_t is a winning price at time t . B, C - predefined constants for budget and cost-per-click (CPC).

Autobidding Problem: Dual Problem

Apply relaxation to the $x_t \in [0, 1]$:

$$\begin{array}{ll}\min_{p,q,r} & B \cdot p + \sum_{t=1}^T r_t \\ \text{s.t.} & wp_t \cdot p + (wp_t - CTR_t \cdot C) q + r_t \geq v_t, \quad \forall t \in [1, T] \\ \text{where} & p \geq 0, \\ & q \geq 0, \\ & r_t \geq 0, \quad \forall t \\ & v_t = CTR_t \cdot CVR_t, \quad \forall t.\end{array}$$

Where p, q corresponds to the Budget and CPC constraint.

Autobidding Formula Derivation

Assume the optimal solution for the primal problem (LP1) is $x_t^*, \forall t = 1, \dots, n$, and the optimal solution of the corresponding dual problem is $p^*, q^*, \{r_t^* \mid i = 1, \dots, n\}$.

According to the theorem of complementary slackness, we obtain:

$$x_t^* \cdot (v_t - wp_t \cdot p - (wp_t - CTR_t \cdot C)q - r_t) = 0, \quad \forall t \quad (1)$$

$$(x_t^* - 1) \cdot r_t^* = 0, \quad \forall t \quad (2)$$

We set

$$bid_t^* = \frac{1}{p^* + q^*} CTR_t \cdot CVR_t + \frac{q^*}{p^* + q^*} C \cdot CTR_t, \quad (3)$$

then we transform Eq. 1:

$$x_t^* \cdot ((bid_t^* - wp_t)(p^* + q^*) - r_t^*) = 0, \quad \forall i \quad (4)$$

Autobidding Formula Derivation (cont.)

1. According to Eq. 4, if the campaign wins opportunity t , i.e., $x_t^* > 0$, then $(bid_t^* - wp_t)(p^* + q^*) - r_t^* = 0$. Meanwhile, $p^* \geq 0$, $q^* \geq 0$, $r_t^* \geq 0$, so $bid_t^* \geq wp_t$.
2. If the campaign loses opportunity t , i.e., $x_t^* = 0$, we could deduce from Eq. 2 that $r_t^* = 0$. Therefore, according to the dual constraint, we obtain $(bid_t^* - wp_t)(p^* + q^*) \leq 0$, namely $bid_t^* \leq wp_t$.

This shows that the bid formula correctly captures the optimal bidding strategy: bid above the winning price when winning, and below when losing. ■

Autobidding Formula

$$bid_t^* = \frac{1}{p^* + q^*} CTR_t \cdot CVR_t + \frac{q^*}{p^* + q^*} C \cdot CTR_t, \quad (5)$$

Where p, q are optimal dual variables for Budget and CPC constraints correspondingly.

More General setting

$$\begin{aligned} & \underset{x_i}{\text{maximize}} && \sum_i v_i x_i \\ & \text{s.t.} && \sum_i c_i x_i \leq B \\ & && \frac{\sum_i c_{ij} x_i}{\sum_i p_{ij} x_i} \leq k_j, \forall j \\ & && x_i \leq 1, \forall i \\ & && x_i \geq 0, \forall i \end{aligned}$$

Theorem

For a sub-problem at each time-step t , the optimal action sequence for the bidding agent is to modify the current $\mathbf{w}_t = (\mathbf{p}, \mathbf{q})$ to the optimal $\mathbf{w}_t^ = (\mathbf{p}^*, \mathbf{q}^*)$, and keep it fixed at the following time steps. [1]*

Online Autobidding

How to update bids without knowing the future

Control-Theoretic Approaches

Let e_t be the deviation of the actual trajectory from the target. Then

$$b_t = b_t^{\text{base}} m_t,$$

$$m_t = m_{t-1} + K_P e_t + K_I \sum_{\tau=1}^t e_{\tau} + K_D (e_t - e_{t-1}).$$

[7]

Methods differ in how they adapt

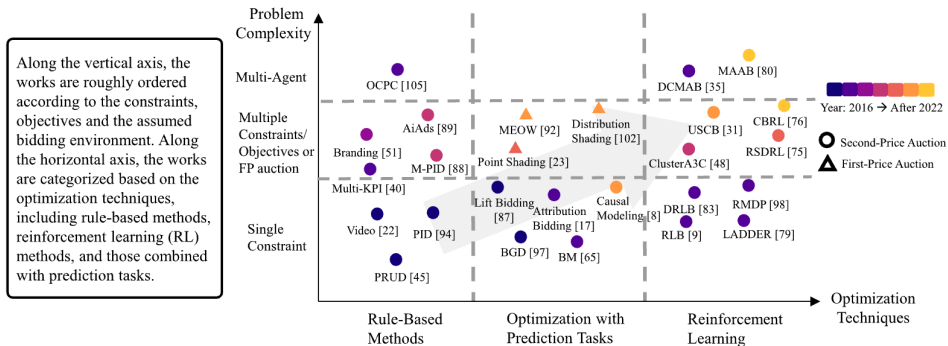


Fig. 1. The overall trend for bid optimization of real-time bidding display advertising.

Optimisation, control, and reinforcement learning address the same decision problem with different information and assumptions. [3]

Reinforcement Learning

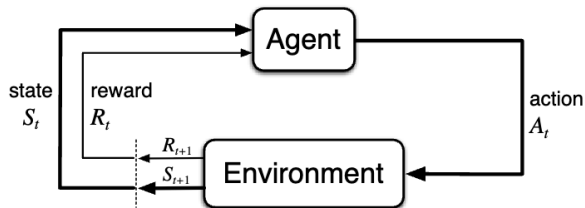


Figure 3.1: The agent–environment interaction in reinforcement learning.

The agent observes campaign state, selects an action, and receives the next state and reward from the environment. [6]

MDP Formulation for Autobidding (RTB)

State s_t

- remaining time;
- remaining budget;
- current KPI status;
- bid-request features;
- spend history.

Action and reward

- bid or multiplier;
- clicks, conversions, value;
- penalty for KPI violation;
- penalty for early depletion.

A useful Markov state must contain enough information to predict the future effect of the action.

Robustness and evaluation

The algorithm must survive model error and an honest benchmark

Prediction error changes the bid directly

$$b_t \propto \widehat{\text{CTR}}_t \widehat{\text{CVR}}_t.$$

- Overestimation causes overpayment and early budget depletion.
- Underestimation loses useful inventory.
- Distribution shift breaks calibration and the controller together.
- Optimising one metric may degrade the others.

The prediction model, bidding policy, and feedback loop must be validated as one system.

RobustBid: Handling Noisy Predictions

$$\begin{aligned} \max_{0 \leq x_t \leq 1} \min_{\mathbf{a} \in \mathcal{U}} & \sum_{t=1}^T x_t \mathbf{a}_t \text{CVR}_t \\ \text{s.t.} & \sum_{t=1}^T x_t w p_t \leq B, \\ & \sum_{t=1}^T x_t w p_t \leq C \sum_{t=1}^T x_t \mathbf{a}_t. \end{aligned}$$

$$\mathcal{U} = \{\mathbf{a} : L(\widehat{\mathbf{CTR}}, \mathbf{a}) \leq \varepsilon\}.$$

The robust correction protects the bid against admissible prediction errors. [5]

RobustBid adds a closed-form correction

For the MSE uncertainty set

$$\mathcal{U}_{\text{mse}} = \left\{ \mathbf{a} \in [0, 1]^T : \frac{1}{2} \left\| \widehat{\mathbf{CTR}} - \mathbf{a} \right\|_2^2 \leq \varepsilon_{\mathbf{a}} \right\}, \quad \alpha = \sqrt{2\varepsilon_{\mathbf{a}}},$$

the CTR-robust optimal bid is the nominal dual bid plus an explicit correction:

$$b_t^{\text{RB}} = \frac{\widehat{\text{CTR}}_t \text{CVR}_t + q^* \widehat{\text{CCTR}}_t}{p^* + q^*} + \delta_t^{\text{ctr}},$$
$$\delta_t^{\text{ctr}} = -\frac{\alpha}{p^* + q^*} \left(\frac{Cq^*}{\sqrt{|\mathcal{T}|}} + \frac{\text{CVR}_t^2}{\sqrt{\sum_{\ell \in \mathcal{T}} \text{CVR}_\ell^2}} \right), \quad t \in \mathcal{T},$$

and $\delta_t^{\text{ctr}} = 0$ outside $\mathcal{T}$.

Larger uncertainty yields a more negative correction; p^*, q^* are the budget and CPC dual variables. [5]

DenoiseBid optimises latent outcomes

$\mathcal{O} = \{(\widehat{\text{CTR}}_t, \widehat{\text{CVR}}_t)\}_{t=1}^T$ are noisy outputs; CTR_t and CVR_t remain latent.

$$\begin{aligned} \max_{0 \leq x_t \leq 1} \quad & \mathbb{E} \left[\sum_{t=1}^T x_t \text{CTR}_t \text{CVR}_t \mid \mathcal{O} \right] \\ \text{s.t.} \quad & \sum_{t=1}^T x_t \text{wp}_t \leq B, \\ & \mathbb{E} \left[\frac{\sum_{t=1}^T x_t \text{wp}_t}{\sum_{t=1}^T x_t \text{CTR}_t} \mid \mathcal{O} \right] \leq C. \end{aligned}$$

A first-order approximation converts the last constraint to

$$\sum_{t=1}^T x_t (\text{wp}_t - C \mathbb{E}[\text{CTR}_t \mid \mathcal{O}]) \leq 0,$$

yielding an LP in posterior moments. [9]

DenoiseBid needs two posterior means

Define the posterior conversion value and click probability

$$v_t = \mathbb{E}[\text{CTR}_t \text{CVR}_t \mid \mathcal{O}], \quad c_t = \mathbb{E}[\text{CTR}_t \mid \mathcal{O}].$$

Lagrangian duality and complementary slackness yield the closed-form bid

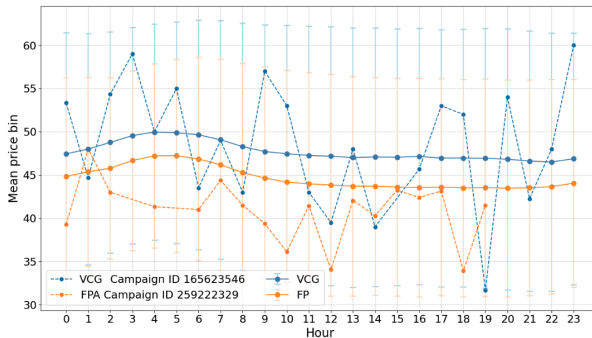
$$\boxed{b_t^{\text{DB}} = \frac{v_t}{p^* + q^*} + \frac{q^*}{p^* + q^*} C c_t} = \frac{\mathbb{E}[\text{CTR}_t \text{CVR}_t \mid \mathcal{O}] + q^* C \mathbb{E}[\text{CTR}_t \mid \mathcal{O}]}{p^* + q^*}.$$

The required moments follow from Bayes' rule:

$$\mathbb{E}[X \mid \mathcal{O}] = \frac{\int X p(\mathcal{O} \mid X) p(X) dX}{\int p(\mathcal{O} \mid X) p(X) dX}, \quad X \in \{\text{CTR}, \text{CTR} \cdot \text{CVR}\}.$$

Posterior means absorb uncertainty; the online bid remains explicit and inexpensive. [9]

BAT: Benchmark for Auto-bidding



BAT formalises reproducible evaluation on standardised tasks and metrics. [2]

Autobidding Arena: Compare Algorithms

	AWR	eCPM	Total clicks	eCPC, CPC=0.5	Total conversions	eCPA, CPA=0.05	RMSE
PID	0.686	297.10	14776	0.59	963	9.28	1586.02
ORTB1	0.546	421.04	15511	0.64	950	10.64	1870.59
Linear	0.565	396.52	14507	0.64	938	10.07	1772.65
FAB	0.691	289.35	14612	0.58	920	9.58	1457.33
MPID	0.926	217.27	15239	0.64	861	11.60	1050.71
OPT	0.799	195.21	15468	0.47	807	9.22	1317.98
BROI	0.508	421.15	15090	0.66	734	14.21	1889.09
USCB	0.691	229.85	13176	0.54	709	10.88	1474.63
RLB	0.406	513.82	12394	0.83	682	16.25	2187.48
CBRL	0.960	139.95	15554	0.42	667	10.32	960.89
CostMax	0.635	328.36	14550	0.70	666	15.89	1525.25
DRLB	0.440	478.39	12795	0.79	664	16.17	2195.81
Const	0.666	313.15	14927	0.68	660	16.06	1584.01
ORTB2	0.427	488.79	11163	0.91	637	17.00	2164.45
Mystique	0.392	538.92	10714	0.94	632	16.22	2014.30
Fb	0.521	383.82	13703	0.71	621	16.35	1619.31
Random	0.387	538.70	8350	1.23	438	24.91	1898.00

Tuning an algorithm for one target metric can degrade other campaign characteristics. [4]

Autobidding Arena: Compare Algorithms



Tuning an algorithm for one target metric can degrade other campaign characteristics. [4]

Summary

1. **Auction:** the bid turns predicted value into a purchase threshold.
2. **Budget and KPI:** dual variables price the constraints.
3. **Feedback:** PID, dual updates, or RL adapt the policy online.
4. **Robustness:** prediction errors belong in the formulation and tests.

Main takeaway

The best bidding algorithm is the one whose assumptions match the data, constraints, and safety loop of the actual advertising system.

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